

Building Equivalent Ratio Tables

Answer Key

Grade 6

Quick Answers

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|---|--|
| 1. flour 4 = 6, flour 6 = 9, flour 8 = 12 | 8. 20 |
| 2. 2 cars = 8, 3 cars = 12, 5 cars = 20 | 9. one cup of rice = 2.5, 7 cups of rice = 17.5 |
| 3. one notebook = 2, 5 notebooks = 10 | 10. second value for 10 = 15, first value for 18 = |
| 4. 2 baskets = 5, 8 baskets = 20, 12 baskets = 30 | 12 |
| 5. 12 | 11. Machine A, one minute = 45, Machine B, one minute = 50, faster machine = < |
| 6. one ticket = 9, 6 tickets = 54 | 12. 6 cups of juice = 10, 9 cups of juice = 15, juice in 24 cups of punch = 9 |
| 7. 9 red = 12, 15 red = 20 | |
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Curriculum

Level: Grade 6

6.RP.A.2 – problems 3, 6, 9, 11

6.RP.A.3 – problems 1, 2, 4, 5, 7, 8, 10, 12

Difficulty 1–4 of 5 across 27 tagged checks.

- 1.** Flour to milk is 2 to 3. Complete the milk row for 4, 6, and 8 cups of flour.

Solution: Each column multiplies the base column (2, 3) by the same number: $4 = 2 \times 2$, $6 = 2 \times 3$, $8 = 2 \times 4$. Multiply the milk value 3 by the same multipliers: 3×2 , 3×3 , 3×4 , giving

,	6
,	9
,	12

- 2.** Wheels to cars is 4 to 1. Complete the wheels row for 2, 3, and 5 cars.

Solution: One car has 4 wheels, so multiply the number of cars by 4: 2×4 , 3×4 , 5×4 , giving

,	8
,	12
,	20

- 3.** 3 notebooks cost 6 dollars. Find the cost of one notebook, then of 5.

Solution: Divide both quantities by 3 to build the unit column: $6 \div 3 = 2$ dollars for one notebook.

2

Then multiply the unit cost by 5: $2 \times 5 = 10$ dollars.

10

4. 4 baskets hold 10 melons. Complete the melon row for 2, 8, and 12 baskets.

Solution: For 2 baskets, divide both quantities by 2: $10 \div 2 = 5$.

5

For 8 baskets, multiply the base column by 2: $10 \times 2 = 20$.

20

For 12 baskets, multiply by 3: $10 \times 3 = 30$.

30

5. Find the missing value: $\frac{3}{7} = \frac{?}{28}$.

Solution: The denominator was multiplied by 4, because $7 \times 4 = 28$. Multiply the numerator by the same 4: $3 \times 4 = 12$.

12

6. 4 tickets cost 36 dollars. Find the cost for one person, then for 6.

Solution: Unit column: divide both quantities by 4, so one ticket costs $36 \div 4 = 9$ dollars.

9

Six tickets: $9 \times 6 = 54$ dollars.

54

7. Red to white is 6 to 8. Complete the white row for 9 and 15 parts red.

Solution: From 6 to 9 the multiplier is $9 \div 6 = 1.5$, so white is $8 \times 1.5 = 12$.

12

From 6 to 15 the multiplier is $15 \div 6 = 2.5$, so white is $8 \times 2.5 = 20$.

20

8. Find the missing value: $\frac{9}{12} = \frac{15}{?}$.

Solution: The numerator was scaled by $15 \div 9 = \frac{5}{3}$, so the denominator is $12 \times \frac{5}{3} = 20$.
Check: $\frac{15}{20}$ and $\frac{9}{12}$ both simplify to $\frac{3}{4}$.

20

9. Water to rice is 10 to 4. Complete the water row for one cup and for 7 cups of rice.

Solution: Unit column: divide both by 4, so one cup of rice takes $10 \div 4 = 2.5$ cups of water.

2.5

Seven cups: $2.5 \times 7 = 17.5$ cups of water.

17.5

10. Every row shows the ratio 12 to 8. Fill in the missing first quantity (paired with 10) and the missing second quantity (paired with 18).

Solution: The ratio $\frac{12}{8}$ simplifies to $\frac{3}{2}$, so the first quantity is always $\frac{3}{2}$ of the second.

Paired with 10: first quantity = $\frac{3}{2} \times 10 = 15$.

15

Paired with 18: second quantity = $18 \div \frac{3}{2} = 12$.

12

11. Machine A: 180 pages in 4 minutes. Machine B: 250 pages in 5 minutes. Build one-minute columns and decide which is faster.

Solution: (a) Machine A: divide both by 4: $180 \div 4 = 45$ pages per minute.

45

(b) Machine B: divide both by 5: $250 \div 5 = 50$ pages per minute.

50

(c) Comparing unit rates, $45 < 50$, so Machine B is faster.

$45 < 50$, Machine B

12. Juice to soda is 3 to 5. Complete the soda row for 6 and 9 cups of juice, then find the juice needed for 24 cups of punch in total.

Solution: For 6 cups of juice the multiplier is 2, so soda is $5 \times 2 = 10$.

10

For 9 cups of juice the multiplier is 3, so soda is $5 \times 3 = 15$.

15

For the last question, one batch of $3 + 5 = 8$ total cups contains 3 cups of juice. To reach 24 total cups, multiply the whole batch by $24 \div 8 = 3$: juice = $3 \times 3 = 9$ cups.

9